METRIC

MIL-PRF-8565/11D
14 July 2009
SUPERSEDING
MIL-B-8565/11C
31 March 1993

PERFORMANCE SPECIFICATION SHEET

# BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 10-AMPERE-HOUR

Reactivated after 14 July 2009 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery covered by this specification sheet shall be D8565/11-1 for figure 1. D8565/11-2 is cancelled without supersession.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1 of MIL-PRF-8565/11. The weight of the battery shall be not greater than 15.8 kilograms (34.8 pounds) (see 3.8.3 and 4.5.3 of MIL-PRF-8565).

3. Capacity. The rated capacity of the battery shall be not less than 10 ampere-hours (Ah), based on its ability to conform to the life test final discharge (10.0 Ah/1 Hr/24°C/18.0 V). See 3.8.6, 4.5.7 and 6.10 of MIL-PRF-8565.

4. Service life. The battery's service life (useful life) in aircraft shall be not less than 3 years.

5. Qualification inspection and conformance inspection. Table I lists the required inspections.



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6. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

6.1 The following paragraphs are not applicable:
3.5.10 Heaters.
3.6.5 Marking for batteries with heaters.
3.8.7 Constant voltage discharge.
3.8.17 Cold and hot temperature charge.
4.5.8 Constant voltage discharge.
4.5.29 Cold and hot temperature charge test.

6.2 Modify the following paragraphs:

3.5.5 Receptacles. After the first sentence, insert: "For D8565/11-1, the female receptacle (front view) shall be dimensionally equal to MS3509, style 1. The receptacle may be integrally molded in the container."

3.6.4 Battery caution marking. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "Marking for Label 2 shown on figure 1 of MIL-PRF-8565/11 shall show the following for battery type D8565/11-1, with lettering height not less than 6.35mm (0.25 inch). The marking shall be in white or black print."

MAINTENANCE-FREE BATTERY

DO NOT REMOVE COVER

THIS BATTERY IS TO BE REMOVED FROM
SERVICE 3 YEARS FROM DATE FIRST PLACED
IN SERVICE
FOR ARMY, CONSULT MIM UH-60-MIM-05-007
FOR SERVICE LIFE EXTENSION
AUTHORIZATIONS.

CAUTION!

DO NOT USE EXCESSIVE TORQUE

ON HOLD-DOWN STRAP BOLTS"

4.5.10 Cycling test. In the third sentence, delete "2.0-hour charge" and substitute "4.0-hour charge".

4.5.12 Charge and discharge test at low temperature. In Step b., Delete "-18°C (0°F)" and substitute "-12°C (10°F)". In steps c and f, delete "1.0 C-rate to a cutoff voltage of 18 volts" and substitute "65 amperes to a cutoff voltage of 14.4 volts".



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6.3 Add the following paragraphs:

"3.6.7 Hold-down hooks. A circular area of 8.5 +/-1 mm radius concentric with each mounting hole of the hold-down hooks shall be left as bare uncoated metal."

4.5.28 Output performance test. Add the following:

"4.5.28.1 Fifteen minute discharge at 27ºC. The battery shall be charged as specified in accordance with 4.4.2 and stored under the specified temperature and storage conditions (see 4.4.1 of MIL-PRF-8565). The battery shall then be discharged at a rate specified in accordance with requirement (6) of table III of MIL-PRF-8565/11, a 90-ampere discharge for 1 minute followed immediately by a 25-ampere discharge for 14 minutes. The batteries shall maintain a terminal potential of 10.0 volts or greater throughout the total 15-minute duration of the two discharges. The battery shall meet the requirement of 3.8.6 of MIL-PRF-8565.

4.5.28.2 Three minute discharge at -12ºC. The battery shall be charged as specified in accordance with 4.4.2 of MIL-PRF-8565 and stored under the specified temperature and storage conditions (see 4.4.1 of MIL-PRF-8565). The battery shall then be discharged at a rate specified in accordance with requirement (3) of table III of MIL-PRF-8565/11, a 65-ampere discharge to a terminal potential of 14.4 volts. The battery shall meet the requirement of 3.8.6 of MIL-PRF-8565.

4.5.28.3 One minute discharge at -30ºC. The battery shall be charged as specified in accordance with 4.4.2 of MIL-PRF-8565 and stored under the specified temperature and storage conditions (see 4.4.1 of MIL-PRF-8565). The battery shall then be discharged at a rate specified in accordance with requirement (4) of table III of MIL-PRF-8565/11, a 65-ampere discharge to a terminal potential of 14.4 volts. The battery shall meet the requirement of 3.8.6 of MIL-PRF-8565."



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TABLE I. Qualification inspection and conformance inspection of batteries.

| Test number | Examinations and tests | Qualification inspection | | | | | Conformance inspection | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Sample number | | | Requirement paragraph 1/ | Test paragraph 1/ | Group A sample 1 100% | Group B sample number | |
| | | 1 | 2 | 3 | | | | 1 | 2 |
| 1 | INCOMING INSPECTION | X | X | X | 3.8.1(G) | 4.5.1(G) | X | X | X |
| 2 | VISUAL AND MECHANICAL EXAMINATION | X | X | X | 3.5 (G), 3.6 (G), 3.8.2 (G) table II (G) | 4.5.2 (G) | X | X | X |
| 3 | DIMENSIONS AND WEIGHT | X | X | X | Reqmt 2 (N), 3.8.3 (G) | 4.5.3 (G) | X 2/ | X 2/ | |
| 4 | DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST | X | X | X | 3.7 (G), 3.8.4 (G) | 4.5.4 (G) | X 2/ | X | X |
| 5 | STRENGTH OF VENT TUBES TEST | X | X | X | 3.7 (G), 3.8.5 (G) | 4.5.5 (G) | X 2/ | X | X |
| 6 | ELECTRICAL TEST PREPARATION | X | X | X | -- | 4.5.6 (G) | | X | X |
| 7 | CAPACITY DISCHARGE | X | X | X | Reqmt 3 (N), 3.8.6 (G), Reqmt 1 of table III (S) | 4.5.7 (G) | X | X | X |
| 8 | STRENGTH OF RECEPTACLE TEST | | | X | 3.7 (G), 3.8.8 (G) | 4.5.9 (G) | | | X |
| 9 | CYCLING TEST | X | | | 3.8.9 (G), Reqmt 1 of table III (S) | 4.5.10 (G) | | X | |
| 10 | CAPACITY DISCHARGE AT 49 °C (120 °F) | | X | | 3.8.6 (G), Reqmt 2 of table III (S) | 4.5.11 (G) | | | |
| 11 | CHARGE AND DISCHARGE TEST AT LOW TEMPERATURE | | X | | 3.8.6 (G), Reqmt 3 of table III (S) | 4.5.12 (G) | | | X |
| 12 | DISCHARGE WHILE INVERTED | | X | | 3.5.9 (G), Reqmt 5 of table III (G) | 4.5.13 (G) | | | X |
| 13 | ALTITUDE TEST | | X | | 3.7 (G), 3.5.9 (G), 3.8.10 (G) | 4.5.14 (G) | | | X |
| 14 | SHOCK TEST (BASIC DESIGN) | | | X | 3.7 (G), 3.8.10 (G) | 4.5.15 (G) | | | |
| 15 | TEMPERATURE SHOCK TEST | X | | | 3.7 (G) 3.8.10 (G) | 4.5.16 (G) | | X | |
| 16 | FIFTEEN-MINUTE DISCHARGE | | X | | 3.8.6 (G) | 4.5.28.1 (N) | X | | X |
| 17 | THREE-MINUTE DISCHARGE | | X | | 3.8.6 (G) | 4.5.28.2 (N) | | | |



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TABLE I. Qualification inspection and conformance inspection of batteries – Continued.

| Test number | Examinations and tests | Qualification inspection | | | | | Conformance inspection | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Sample number | | | Requirement paragraph 1/ | Test paragraph 1/ | Group A sample 1 100% | Group B sample number | |
| | | 1 | 2 | 3 | | | | 1 | 2 |
| 18 | ONE-MINUTE DISCHARGE | | | X | 3.8.6 (G) | 4.5.28.3 (N) | | | X |
| 19 | TEMPERATURE RISE AND FLOAT TEST | | X | | 3.7 (G), 3.8.11 (G), Reqmt 1 of table III (S) | 4.5.17 (G) | | | X |
| 20 | BATTERY GAS EMISSION TEST | | X | | 3.8.10 (G), 3.8.12 (G) | 4.5.18 (G) | | | |
| 21 | VIBRATION TEST | | X | | 3.7 (G), 3.8.10 (G) | 4.5.19 (G) | | | X |
| 22 | HUMIDITY TEST | | X | | 3.7 (G), 3.8.10 (G) | 4.5.20 (G) | | | X |
| 23 | SALT FOG TEST | X | | | 3.7 (G) 3.8.10 (G) | 4.5.21 (G) | | X | |
| 24 | GROUND STORAGE TEST | | | X 3/ | 3.8.13.1 (G) | 4.5.22 (G) | | | |
| 25 | DEEP DISCHARGE RECOVERY TEST | | | X | 3.7 (G), 3.8.13.3 (G) | 4.5.27 (G) | | | |
| 26 | PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST | X | | | 3.7 (G), 3.8.10 (G), Reqmt 1 of table III (S) | 4.5.23 (G) | | | |
| 27 | SHOCK TEST (CRASH SAFETY) | | | X | 3.8.10 (G) 3.8.15 (G) | 4.5.25 (G) | | | |
| 28 | IMPACT RESISTANCE TEST (NONMETALLIC BATTERY CONTAINER) | | 4/ | | 3.8.14 (G) | 4.5.24 (G) | | | |
| 29 | WORMGEAR RETAINING PIN HARDNESS | | 5/ | | 3.5.5 (S) | 4.5.30 (G) | | | |
| 30 | SHELF LIFE TEST | | | X | 3.8.9 (G) 3.8.13.2 (G) | 4.5.26 (G) | | | |
| 31 | FINAL EXAMINATION | X | X | X | 3.7 (G), 3.8.4 (G) | 4.5.31 (G) | | X | X |

1/ The notations in parentheses specify the source of the paragraph. “(G)” means the paragraph as shown in MIL-PRF-8565. “(S)” shows that the paragraph has been modified by this specification sheet. “(N)” indicates that a new paragraph or requirement has been added by MIL-PRF-8565/11.

2/ See sample information in test paragraph.

3/ The manufacturer requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.



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4/ For nonmetallic battery container designs, two outer nonmetallic battery containers shall be furnished for inspection.

5/ Five wormgear retaining pins shall be furnished for inspection.

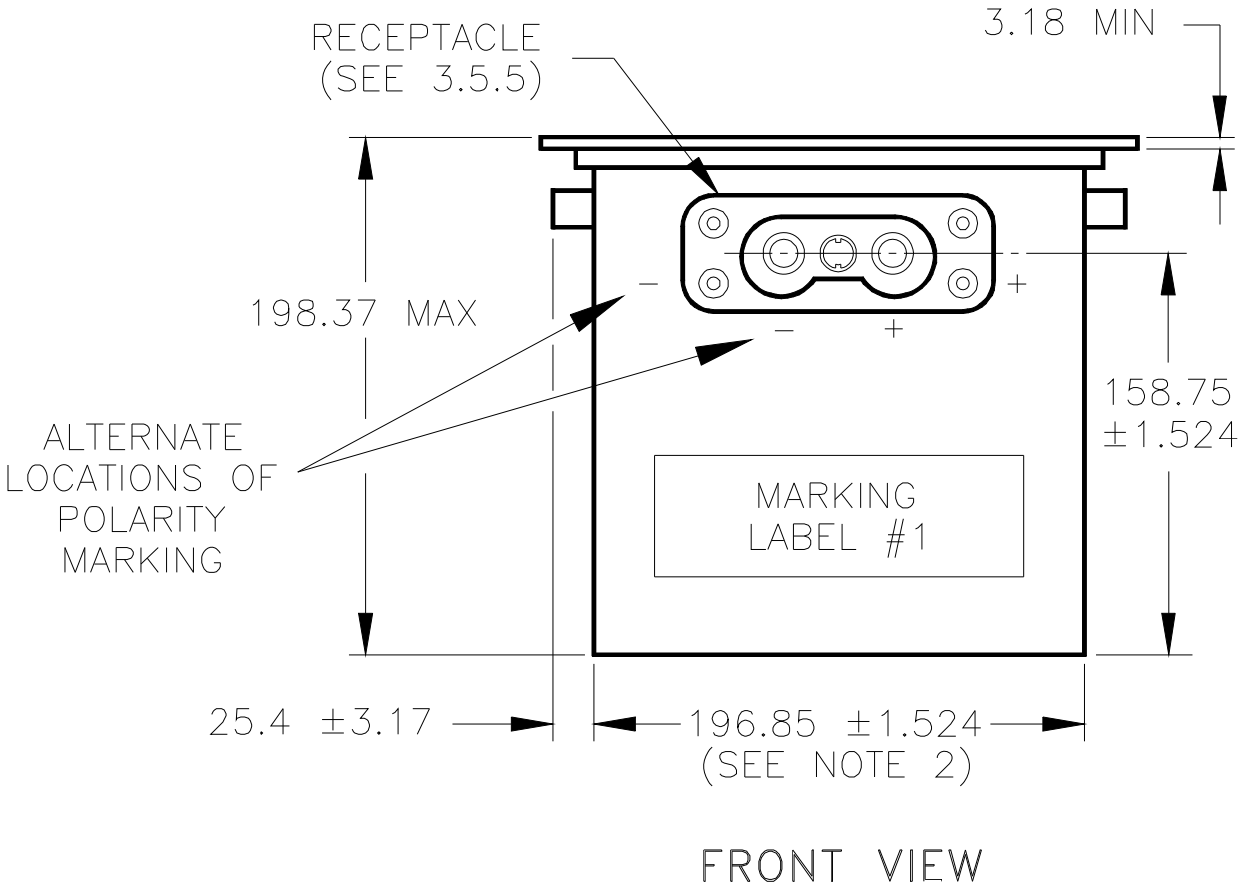
TABLE III. Capacity performance requirements. Delete the table and insert the new table III.

TABLE III. Capacity performance requirements.

| Requirement Number | Discharge cutoff volts & temperature | Rate of discharge factor | Minimum time to cutoff voltage | |
|---|---|---|---|---|
| (1) | 18.0 volts at 24 ± 3°C | 1.0C | 1 hour | |
| (2) | 18.0 volts at 49 ± 3°C | 1.0C | 1.1 hour | |
| (3) | 14.4 volts at -12 ± 3°C | 6.5C | 3 minutes | |
| (4) | 14.4 volts at -30 ± 3°C | 6.5C | 1 minute | |
| (5) | 14.0 volts at 27 ± 3°C | 6.5C | 5 minutes | |
| (6) | 10.0 volts at 27 ± 3°C | 9.0C<br>2.5C | 1 minute<br>14 minutes | 15 minutes |



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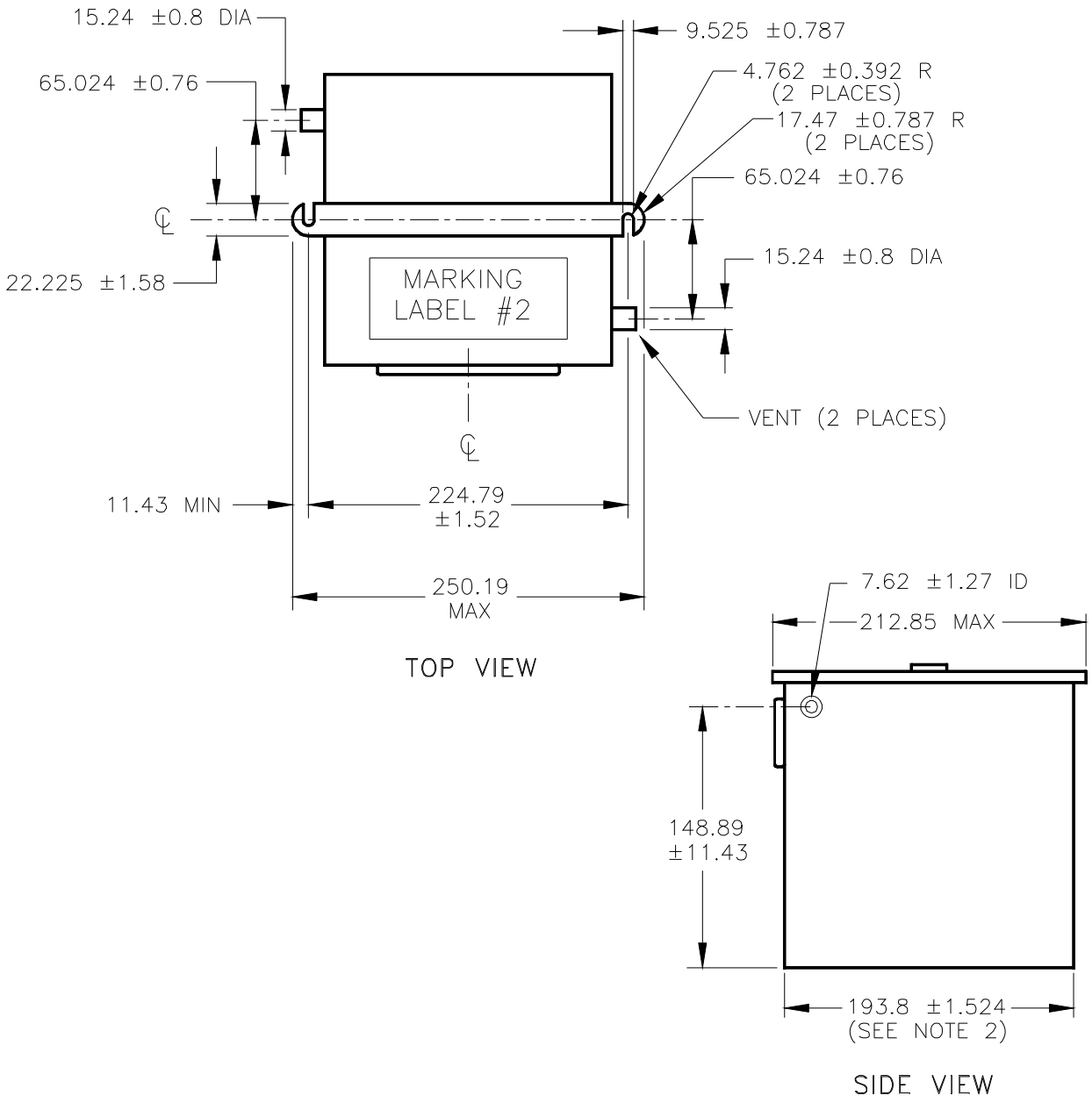

NOTES:

1. All dimensions are in millimeters.
2. Allow additional 10 mm width for cover.
3. Abbreviations used on this figure are maximum (MAX) and minimum (MIN).

FIGURE 1. PIN D8565/11-1 dimensions.



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NOTES:

1. All dimensions are in millimeters.
2. Allow additional 10 mm length for cover.
3. Abbreviations used on this figure are diameter (DIA), inside diameter (ID) maximum (MAX), minimum (MIN) and radius (R).
4. Leave an uncoated area of 8.5 +/-1 mm R concentric with the mounting hole in 2 places.

FIGURE 1. PIN D8565/11-1 dimensions - Continued.



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6.19 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy – AS
Air Force – 84

Review activity:
Air Force – 99
DLA – GS

Preparing activity:
Navy - AS
(Project 6140-2008-006)

Agent:
Navy – SH

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